

Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

Objective :

1. To develop an Embedded C program to implement serial communication using UART with a serial interrupt in the 8051 microcontroller.
2. To receive keyboard input through UART and process the received data using the serial interrupt.

Code:

```
#include <reg51.h>
```

```
#include <stdio.h>
```

```
void serial_ISR(void) interrupt 4
```

```
{
```

```
char ch;
```

```
if (RI)
```

```
{
```

```
RI = 0; // Clear receive interrupt flag
```

```
ch = SBUF; // Read received character
```

```
printf("\n>>> INTERRUPT RECEIVED: ");
```

```
printf("%c\n", ch);
```

```
}
```

```
}
```

```
void delay(void)
```

```
{
```

```

unsigned int i;

for(i = 0; i < 50000; i++); // Delay for visibility
}

void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
    TMOD = 0x20; // Timer1 mode 2 (auto-reload)
    TH1 = 0xFD; // Baud rate 9600
    TR1 = 1; // Start Timer1

    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)

    TI = 1; // Enable printf

    while(1)
    {
        printf("Hello World\n");
        delay();
    }
}

```

Output:

